NULL CALIBRATION FOR MULTIPLE-CHOICE EVALUATION: WHEN "ABOVE CHANCE" FAILS THE INPUT-BLIND FLOOR

Anonymous authors
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ABSTRACT

Multiple-choice scores are commonly interpreted against chance. We argue that this comparison is often insufficient: before an arm is compared with another arm, its score should be tested against the construct's best constant, input-blind predictor. This floor is a necessary, not sufficient, validity condition. Across ten target constructs and one negative control, the floor is not chance and, for content scoring, is under-specified by the tie convention, length unit, and tokenizer; when option count varies, even "chance" is ambiguous. The floor must itself be calibrated: because it is a maximum over k noisy label marginals, it is upward biased even under exactly uniform labels, and the calibrating null must be realisable by the construct — a test we initially failed on our own largest benchmark. Under a null restricted to each item's legal letters, only two of the eight letter constructs (MMLU, BoolQ, both fixed-k) have floors a balanced null could not produce ($p < 10^{-5}$); on the remaining six the floor lies inside the estimator's own sampling noise ($p = 0.083$–0.853), so those constructs establish that chance is the wrong *reference* without establishing that the floor differs from it in magnitude. On MMLU-Pro, all 21 evaluated cells are powered at the scale of the reference effect. Among 17 designated damaged cells in four model families, the null choice alone reverses the reading: damaged arms appear above a chance line while only two clear their best-constant floor, and the honest aggregate is 15/17 at or below the floor. Held to a single evidentiary standard — a paired bootstrap interval excluding zero on *both* sides, not only the floor side — the flip is 3/12 above chance versus 1/12 above the floor, about a third of the size the asymmetric point-estimate comparison suggests. Across five smaller benchmarks, the designated set gives 9/85 cells above their floor against 45/85 above chance, the same wrong-null flip, while a mandatory power analysis explains why most per-benchmark significance tests are inconclusive. We then adapt a complementary, arm-conditional permutation null for a different question: whether one arm's predictions contain item-level information. Its statistic is the numerator of Cohen's κ evaluated within option-count strata, so it is exactly zero for every pure constant emitter independent of collapse letter; our contribution is the stratification and its use as a pre-comparison gate, not the statistic. We use it to re-judge 27 existing cells. The re-analysis withdraws one above-floor capability label, dissolves both below-floor competence labels, and exposes a weak intact-family anchor. Finally, full-fp32 evaluation removes every bf16 exact tie and changes 18.03% of letter decisions without improving accuracy, ruling out a numerical-tie explanation for the measurement failure. Together, the results motivate a simple reporting rule: calibrate each reported construct to its explicit input-blind null before interpreting arm differences.

1 INTRODUCTION

A recent multilingual evaluation reported an awkward pattern: every tested model was above chance, yet none beat the benchmark's majority-class baseline (Arčon et al., 2026). The authors found the flip because they happened to print both reference lines. This is not merely a curiosity about one newly constructed benchmark. A systematic construct-validity review of 445 language-model benchmarks offers 27 actionable checklist items, but none asks authors to report a null, a chance level, or a constant predictor (Bean et al., 2025). The missing item is operational: *before comparing model arms, test whether the reported construct clears its own input-blind floor*.

For a letter-valued multiple-choice construct, the floor is the most accurate constant letter under the benchmark's empirical gold marginal. For content likelihood, an analogous input-blind heuristic can exploit option length. These references need not equal nominal chance. More importantly, an arm can look "above chance" while carrying no more usable signal through the measured interface than a constant predictor. This distinction is especially consequential under structural damage, where option-token selection bias (Zheng et al., 2024) can dominate the argmax while task-relevant content knowledge remains partly available.

We study null calibration as a *measurement protocol*, not as a new multiple-choice interface or a new debiasing algorithm. Majority-class baselines for MCQA already exist (Balepur et al., 2024); letter-versus-content formulations are established (Gu et al., 2025; Cho et al., 2026); and prior work has shown that constant outputs can game automatic judges (Zheng et al., 2025). Our contribution is to make the construct-appropriate input-blind null a pre-comparison gate, quantify the degrees of freedom hidden inside that null, and separate two questions that require different references:

1. **Is the interface valid for arm comparison?** Use an arm-independent best-constant floor.
2. **Does this particular arm carry item-level information?** Use an arm-conditional permutation null that preserves the arm's prediction marginal.

The distinction prevents two opposite mistakes. Chance can credit a literal constant emitter as competent; conversely, the best-constant floor can label two equally empty emitters differently solely because they collapse onto different letters. The first problem motivates our headline floor. The second motivates read-out v2, for which we prove constant-collapse invariance algebraically.

Our empirical evidence uses ten target constructs plus Winogrande as a negative control, four base-model families, letter and content likelihood interfaces, and structurally damaged arms. The main findings are:

- The null is not "chance," and the floor is itself an estimate that needs its own calibration against a null the construct can *realise* — a test we initially failed on our largest benchmark, where uniform-over-ten-letters assigns gold letters 17% of MMLU-Pro items cannot have. BoolQ's best constant is 0.6217, not 0.50; MMLU-Pro's floor is 0.116606, or 1.1661× naive ten-way chance. We report both gaps and ratios because either alone distorts comparisons across option counts. Against a null restricted to each item's own legal letters (Table 1), only MMLU and BoolQ have floors it could not plausibly produce ($p < 10^{-5}$), and both have a genuinely fixed option count; the six others, MMLU-Pro included, lie inside the estimator's sampling noise ($p = 0.083$–0.853). We therefore rest the quantitative claim on MMLU and BoolQ, reading the rest as showing only that chance is the wrong reference to *report*, not that the floor is far from it. MMLU-Pro at $p = 0.083$ is unresolved, not settled; Section A.1.5 says which rows moved and how.
- Content floors are not dataset constants unless their tie convention, length unit, and tokenizer are fixed. On MMLU-Pro, one tokenizer and one dataset yield floors from 0.125914 to 0.532164 across tie conventions; the latter exceeds the intact model's content score by 32.5 points.
- At MMLU-scale power, 17 designated damaged cells across four families exhibit wrong-null readings: 10/12 non-OLMo cells exceed item-averaged chance (12/12 exceed naive 0.10), yet only 1/12 clears the floor. The honest statement is 15/17 at or below the arm-independent floor, with two exceptions that differ in kind — `qwen3_8b_base/k14` is statistically real but materially negligible, while OLMo-2 `shortgpt16` retains 16 of 32 layers and clears its floor by $5.2\times$ its own half-width. Off MMLU the same set gives 9/85 above floor against 45/85 above chance, and all nine are healed arms retaining at least 14 layers; the accompanying power table is essential.
- The permutation null makes every pure constant emitter score exactly zero, re-sorts rather than merely shrinks the 27-cell read-out, and reveals that an intact Llama-2-7B is itself a weak anchor on MMLU-Pro.
- Full-fp32 evaluation removes 100% of bf16 exact ties and changes 2,532 letter decisions, but does not recover accuracy. The failure is therefore not repaired by numerical precision.

The floor test is deliberately one-directional. A failure disqualifies the score as evidence for an arm comparison, but a pass does not certify that the benchmark measures the intended capability. Feng et al. (2019) construct a dataset where a partial-input baseline is at chance while artifacts remain exploitable; thus clearing a null is *necessary, not sufficient*.

2 RELATED WORK

Input-blind and constant baselines. Balepur et al. (2024) define a majority-class baseline for letter MCQA and recommend stronger-than-chance references; their object is dataset cheatability under a black-box generative setup, whereas ours is a per-arm gate for likelihood-scored constructs. The distinction is operational: their invalid-output analysis imputes either 0.0 or 0.25 on MMLU, while the construct's best-constant letter value is 0.2689. Zheng et al. (2025) use the term "null model" for crafted constant responses that adversarially exploit an LLM judge. Their null is an attack optimized against a judge; ours is a reference derived from benchmark statistics and requires no attacker. The metalinguistic preprint that motivates our opening reports the above-chance/below-majority pattern descriptively on one new benchmark, but does not turn it into a pre-comparison protocol (Arčon et al., 2026).

Selection bias and MC interfaces. Letter preference is not a new observation. Zheng et al. (2024) identify selection and token bias and propose PriDe, which estimates an option-ID prior from permuted contents. PriDe fixes the predictor; null calibration fixes the reference line. Even after prediction debiasing, reporting only against chance can leave the benchmark's gold-marginal floor unreported. We therefore do not reject permutation controls on cost grounds: they are a useful and potentially inexpensive diagnostic. They answer whether predictions should be corrected, while our arm-independent floor answers whether the resulting interface score is comparable across arms.

OLMES standardizes both multiple-choice formulation (MCF) and cloze formulation (CF), then uses the better score per task and model (Gu et al., 2025). This is not a size-keyed rule. Its reference discussion is nevertheless framed around a random baseline rather than a measured label-marginal floor, so the max rule can select an interface without testing whether that interface clears an input-blind reference. Our ARC-Easy result shows a case where CF rescues an at-floor letter read-out; MMLU shows that an interface swap need not do so. Cho et al. (2026) independently study symbols, cloze, and hybrid formats and propose a new question-sensitive score. We do not claim the interface contrast; our method instead gates whichever construct is reported.

Length-based scoring. Length-normalized likelihood can over-correct and its token-based form is tokenizer-dependent (Oostermeijer, 2026). We therefore do not claim generic length sensitivity or tokenizer dependence of the scoring rule. We show that the *induced input-blind floor* inherits these choices: its numerical value changes with tie convention, character versus continuation-token length, and tokenizer. OLMES's observation that tokenizer dependence may be irrelevant when ranking options for one fixed model and tokenizer is compatible with our result; our scope is cross-model comparison of the reference itself.

Nulls as interpretive controls. Our arm-conditional read-out v2 is a chance-corrected agreement statistic: Δ_{perm} equals the numerator of Cohen's κ (Cohen, 1960) computed within option-count strata, and the constant-emitter zero we rely on is that literature's standard property rather than a new result. We claim only the stratification by variable option count and the use of the statistic as a pre-comparison gate.

Control tasks and selectivity can reverse conclusions about which representation layer is informative (Hewitt & Liang, 2019). That is the closest structural precedent: their null randomizes supervision for a probe, while ours removes input dependence from an MC construct. Ding et al. (2021) similarly motivate sensitivity and specificity tests for representation similarity. Finally, Bean et al. (2025) provide the construct-validity vocabulary and a broad benchmark checklist. We position null calibration as the missing operational item, not as the invention of construct validity.

3 NULL-CALIBRATED MEASUREMENT

Let a benchmark contain items $i = 1, \ldots, n$, gold answers y_i, and a scored interface producing predictions $\hat{y}_i$. Its empirical accuracy is

$$\mathrm{acc} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\hat{y}_i = y_i]. \tag{1}$$

3.1 READ-OUT V1: THE ARM-INDEPENDENT INTERFACE FLOOR

For a letter construct with legal labels $\mathcal{L}$, define the input-blind constant family $h_L(x) = L$. The construct floor is

$$f_{\mathrm{const}} = \max_{L \in \mathcal{L}} \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[y_i = L]. \tag{2}$$

The v1 statistic is $\Delta_{\mathrm{floor}} = \mathrm{acc} - f_{\mathrm{const}}$. Because f_{const} depends only on the evaluated item set, every arm is compared to the same reference. This arm independence is essential: otherwise arm-to-arm comparisons can change because the null changes with the arm.

For content likelihood, we use an input-blind longest-option family. Its prediction is determined solely by option length, but the family is not fully specified until a tie convention and length unit are fixed; token-based variants additionally require a tokenizer. Section 3.3 treats these choices as part of the construct definition.

Our decision rule is intentionally asymmetric. An arm that fails to beat the floor cannot support a claim that the interface expresses arm-specific capability above the strongest constant reference. An arm that clears it has passed only a necessary test. It may still exploit other artifacts that no single null enumerates (Feng et al., 2019).

3.2 READ-OUT V2: AN ARM-CONDITIONAL INDEPENDENCE NULL

The v1 floor asks whether an interface is suitable for mutually comparable arm scores. It is not collapse-invariant as a competence statistic: on the full MMLU-Pro item set, always-A, always-E, and always-J contain the same item-level information—none—yet v1 assigns 0.000, -2.111, and -3.807 percentage points respectively because the gold marginals are non-uniform. The separate ten-letter implementation self-test below conditions each letter on items where it is legal.

For the arm-specific question, we preserve the arm's own prediction multiset and remove its alignment with items. MMLU-Pro has variable option count, so permutations are uniform *within* `n_opt` strata s. If n_s is the stratum size and $c^{\mathrm{pred}}_{s,L}$ and $c^{\mathrm{gold}}_{s,L}$ are prediction and gold counts, then

$$\widehat{\mathrm{acc}} = \mathbb{E}[\mathrm{acc} \mid \hat{\boldsymbol{y}} \text{ permuted within } s] = \sum_{s} \frac{1}{n n_s} \sum_{L} c^{\mathrm{pred}}_{s,L} c^{\mathrm{gold}}_{s,L}, \tag{3}$$

and

$$\Delta_{\mathrm{perm}} = \mathrm{acc} - \widehat{\mathrm{acc}}. \tag{4}$$

The statistic is not new; the stratification and its use are. Δ_{perm} is the *numerator* of Cohen's κ evaluated within `n_opt` strata: writing $p_o = \mathrm{acc}$ and $p_e = \widehat{\mathrm{acc}}$, one has $\kappa = (p_o - p_e)/(1 - p_e)$ and hence $\Delta_{\mathrm{perm}} = \kappa\,(1 - p_e)$ identically. We verified the identity on all 27 cells of Table 7; it holds to the printing precision in every row. We therefore claim neither this statistic nor the collapse identity below as new: that chance-corrected agreement vanishes for a constant rater is the textbook property, since $p_o = p_e$ there. Our contribution is (i) stratifying the permutation within variable option count, so letters that are illegal for an item are never credited, and (ii) using the quantity as an *arm-conditional pre-comparison gate* with an explicit materiality bar rather than as an agreement score. We report Δ_{perm} rather than κ to keep it on the same percentage-point scale as the v1 floor, which is what makes the two nulls directly comparable (Table 3).

Option-count-aware chance correction is decades old; only the varying-k stratification is ours. That a chance term should depend on how many options an item offers long predates this work (Bennett et al., 1954; Brennan & Prediger, 1981; Frary, 1988; Brenner & Kliebsch, 1996; Vries et al., 2008), and we claim none of it: not correcting for the number of options, not choosing p_e as a methodological decision, not noticing that κ-family statistics depend on k. Those works assume a single global k; what we are not aware of there is a null in which k *varies item to item* ($\texttt{n\_opt} \in \{3, \ldots, 10\}$, eight strata here), the permutation confined within strata so a letter illegal for an item can never be credited to it. Appendix A.2 states that boundary claim by claim; its footprint is the 36 items below.

Constant-collapse identity. For a pure always-L emitter, $P(\hat{y} = L') = \delta_{L'L}$. Writing $m_{L'}$ for the relevant gold marginal,

$$\widehat{\mathrm{acc}} = \sum_{L'} \delta_{L'L} m_{L'} = m_L = \mathrm{acc}, \qquad \therefore \quad \Delta_{\mathrm{perm}} = 0. \tag{5}$$

This is an algebraic identity for every legal collapse letter, not an empirical regularity, and it is the $\kappa = 0$ property specialised to our stratified estimator. The implementation asserts zero to below 10^{-12} for all ten letters. The loophole "collapse onto a different letter" is therefore closed by construction.

The two nulls must not be conflated, and their ordering is weaker than it first appears. Unstratified, $\sum_L p_L m_L \le \max_L m_L$ places the v2 null at or below the v1 best-constant floor; our within-stratum permutation only gives $\widehat{\mathrm{acc}} \le \sum_s w_s \max_L m_{s,L}$, whose right-hand side *dominates* f_{const} — on MMLU-Pro by exactly 36 items, or 0.299 pp, because the per-stratum argmax is not always-A. The ordering holds empirically in all 27 cells of Table 7 but becomes a theorem only under a regularity condition we can state and not assume, and an $\texttt{n\_opt}$-conditional emitter legal on every item attains $f_{\mathrm{const}} + 0.299$ pp. Appendix A.3 gives the per-stratum argmaxes, why the 27-cell count overstates the evidence, the tightest cell's attainable violation (0.085 pp), and the condition. **Passing v2 must never be described as clearing the paper's floor.**

We estimate uncertainty with 10,000 paired bootstrap resamples, recomputing $\widehat{\mathrm{acc}}$ inside every resample, and verify the decision with 10,000 within-stratum permutations; both use seed 7 and must agree at $\alpha = 0.05$. Significance is not enough at $n = 12032$. We define

$$\texttt{recovery\_fraction} = \frac{\Delta_{\mathrm{perm}}}{\Delta_{\max}}, \tag{6}$$

where $\Delta_{\max}$ is the best alignment achievable by reassigning the observed prediction multiset, and require at least 0.10 times the same-family intact anchor for a material signal. If the intact anchor itself has $\texttt{recovery\_fraction} < 0.10$, relative claims are blocked.

Pre-registration status. The v2 rule, gates, stratification, and materiality constant were committed before re-judging the existing cells and before future unscored milestones. The 27 numbers reported here are nevertheless post-hoc by construction because those cells had already been scored, and the designer had already seen the shape of the collapse defect. We therefore present v2 as a pre-registered rule for future read-outs and a transparent post-hoc re-analysis of the existing cells.

3.3 Four under-specifications of the reference

A content floor is not a dataset constant. Four choices usually left implicit each move it by more than the effects under study, so all four are part of the measured construct and must be printed with the score. *(i) The tie convention:* on MMLU-Pro, one dataset and one tokenizer give 0.125914 under `wrong` and 0.532164 under `credit`, a 40.6-point span whose upper end exceeds the intact base model's own `content_norm` score by 32.5 points, and on ARC-Challenge `credit` alone moves five of six OLMo-2 arms from significantly above the floor to significantly below it. *(ii) The length unit*, characters or continuation tokens: OpenBookQA `credit` is 0.416 versus 0.644, ARC-Challenge 18.60% tied-longest under characters against 50.85% under tokens. *(iii) The tokenizer*, which moves `credit` by up to 10.6 points on four-way tasks and 9.26 on MMLU-Pro, non-monotonically in vocabulary size (Oostermeijer, 2026). *(iv) The meaning of "chance"* when `n_opt` runs from three to ten: naive $1/10$ gives 0.100000 and $\mathrm{mean}(1/\texttt{n\_opt})$ gives 0.110877, so the gap to the 0.116606 always-A floor is either $+1.661$ points and $1.1661\times$ or $+0.573$ points and $1.0517\times$; both must be reported. The four are independent, so a token content floor is fixed only once all four are — a property of (dataset, convention, unit, tokenizer), not of the dataset. The letter floor is exempt from all four: a pure property of the item set's gold labels, invariant across all 15 cross-family arms and bit-identical across all 21 MMLU-Pro cells. Appendix A.1 develops each with Tables 1 and 2.

4 Experimental Setup

Benchmarks and constructs. We evaluate letter constructs on MMLU, ARC-Challenge, ARC-Easy, OpenBookQA, CommonsenseQA, PIQA, MMLU-Pro, and Winogrande as a negative control; BoolQ supplies an additional binary label floor. Content-side longest-option constructs are evaluated on MMLU, MMLU-Pro, and the five non-MMLU evidence tasks, with Winogrande retained only as a control. The target inventory comprises ten constructs plus the control; OpenBookQA content is reported under both character and token units rather than counted twice.

Models and structural damage. The four families are OLMo-2-7B, Llama-2-7B, Llama-3-8B, and Qwen3-8B-Base. OLMo-2 arms are prune-then-heal checkpoints, including `keep8`, `keep10`, `keep12`, `keep14`, and `shortgpt16`. Non-OLMo damage is evaluation-time front-N truncation: the intact base model is loaded and its layer list is replaced by the first N blocks, with no fresh block, optimizer, gradient, healing, or damaged checkpoint. This is a regime confound, not a clean family contrast, and every joint table labels it. Qwen3 has 36 layers whereas the other families have 32; consequently `k8` retains 22.2% versus 25.0% of the stack.

Interfaces and scoring protocol. The letter prompt is the content prompt with the labelled option body inserted before `Answer:`; the letter interface scores the candidate continuations " A", " B", and so forth. The content interface omits the labelled body and scores option text. Scoring is likelihood-based throughout: summed token log-probability over each candidate continuation followed by argmax. There is no sampling, no decoding, and no regular expression over generated text.

All results use `chat_template = False`. The evaluated systems are base language models with no supervised fine-tuning and no reinforcement learning from human feedback, so applying a chat template would be an unfair and non-comparable measurement. We also use `add_bos = 0`, `desc_style = none`, fp32 master weights with bf16-autocast forward, and batch size 48. MMLU-Pro cross-family runs use maximum length 2048; every final cell has zero truncation.

Statistics and integrity. For deterministic per-item null predictions we report 10,000-resample paired-bootstrap intervals and two-sided mid-p values with seed 7, plus exact McNemar tests. V2 additionally uses 10,000 within-stratum permutations. Every cell requires shard indices exactly $\{0, \ldots, 7\}$, expected cardinality, zero duplicate IDs, zero NaNs, zero truncations, and `chat_template is False`; partial merges raise an error.

Power design. The five smaller evidence benchmarks are unable to resolve every effect of interest, so null results are never interpreted without the power table (Table 4, Appendix A.4). MMLU-Pro supplies $n = 12032$ and makes all 21 letter-floor cells capable of detecting MMLU's -1.389-point reference effect; their 95% half-widths range from 0.083 to 0.968 points.

Designated damaged cells, and multiplicity. Every denominator below is over *designated damaged* cells: all five OLMo-2 prune-then-heal arms and all four truncation rungs of each non-OLMo family, giving 17 MMLU-Pro cells and 85 off MMLU, with Winogrande the sole excluded control. The designation is fixed by construction, never by a measured score, and the set is identical in every denominator; Appendix A.4 records that an earlier version narrowed it on MMLU-Pro without saying so. We report per-cell $\alpha = 0.05$ decisions with no family-wise correction, so headline counts are counts of per-cell decisions rather than simultaneously valid claims; the conclusions rest on the aggregate's direction and on the regime-and-depth pattern organising its exceptions, not on one cell.

5 Results and Analysis

5.1 Wrong-null verdict flips at MMLU-scale power

MMLU-Pro provides the cleanest test (per-cell values in Table 5, Appendix A.8). Under $\mathrm{mean}(1/\texttt{n\_opt})$, 10/12 damaged non-OLMo cells read above chance; under naive 0.10, 12/12 do. Yet only 1/12 clears the best-constant floor. Four of the five designated damaged OLMo-2 cells are likewise above either chance line but not above the floor; the fifth, `shortgpt16`, is above both. Across these 17 cells, the reference changes the interpretation; the exact chance-line counts are reported rather than compressed into a universal categorical flip.

Both sides of the flip, held to one standard. The counts above are not symmetric in their evidentiary standard: the floor comparison carries a bootstrap CI, a bootstrap p, and an exact McNemar test, whereas "above chance" was a bare point comparison. Since $\mathrm{mean}(1/\texttt{n\_opt})$ is, like the floor, a deterministic function of the evaluated item set, the matching test needs no new measurement — the paired item bootstrap with the reference recomputed inside each resample. Applying it (10,000 resamples, seed 7) to the same 12 damaged non-OLMo cells:

Reference	Point estimate above	CI_{95} excludes 0
Chance, $\mathrm{mean}(1/\texttt{n\_opt})$	10/12	**3/12**
Best-constant floor	3/12	**1/12**

So the flip survives a symmetric test but is about a third of its advertised size: 3/12 versus 1/12 rather than 10/12 versus 1/12. Seven of the ten cells that read "above chance" were never above chance by any tested standard — their intervals cover zero. This does not weaken the paper's rule, which is that the *reference* must be stated and calibrated; it removes an asymmetry that inflated the apparent effect of choosing the wrong one. We report both columns throughout and hold the two references to the same standard: the criterion is the two-sided 95% interval on *both* sides, matching the verdict rule already used for the floor.

Two qualifications belong with this two-reference comparison, both carried in Appendix A.5. The exact McNemar instrument does not transfer to the chance reference, which is not a $0/1$ predictor; and under Benjamini–Hochberg at $q = 0.05$ — or Bonferroni — *neither* reference retains a single cell (0/12 both sides), so the corrected comparison is undefined rather than 3/12 versus 1/12. What survives correction is the count itself: observing 3 or more rejections out of 12 has binomial probability 0.0196 under the global null. We therefore read this comparison as evidence that the chance side is not uniformly null, not as three simultaneously valid per-cell claims.

The result is strong but not universal, and the two exceptions differ in kind. Fifteen of the 17 cells are at or below the floor. `qwen3_8b_base/k14` is $+0.233$ points above it ($p = 0.0192$, half-width 0.191). V2 confirms a small alignment effect, $+0.267$ points at $p = 0.0066$, but assigns `recovery_fraction`=0.049, only 9.1% of the intact-family anchor and below the 10% materiality bar. It is a real but immaterial exception, not evidence that `k14` retains MMLU-Pro competence. OLMo-2 `shortgpt16` is not borderline: $+3.674$ points at $p = 0.0001$, $5.2\times$ its own half-width. It retains 16 of 32 layers, more than any `keepN` rung, and we read it as a positive result — the arm retaining most of the stack is the one whose read-out survives, so the floor test tracks capability rather than failing everything put to it (Appendix A.4).

The most vivid small-benchmark cases are literal constant emitters: sixteen damaged cross-family cells have accuracy equal to the marginal of the emitted letter to machine precision, and two OpenBookQA cells emit A on every item, score exactly 0.276000, and land on the optimal constant with $\Delta = 0.000$ points and CI95 $[0, 0]$ — yet read 2.6 points above chance 0.25. Off MMLU the designated set gives 9/85 above floor and 45/85 above chance, and the nine are the shallowest-pruned healed arms: all retain at least 14 of 32 layers, while 0/15 OLMo-2 cells at 12 layers or fewer and 0/60 truncate-only cells at any depth clear a floor. Only 7/60 are significantly below because 52/60 are underpowered for the MMLU reference effect. That is a power limit, not a small effect: ARC-Challenge's median damaged effect, -3.840 points, is larger than MMLU's -3.603, but its half-width is 3.92 rather than 1.18 points (Appendix A.6).

5.2 READ-OUT VERSUS RETAINED KNOWLEDGE

A floor-level letter score does not imply that all task-relevant competence is gone. On ARC-Easy, OLMo-2 `keep8` scores 0.2584 on letters, statistically at its 0.266414 letter floor, while `content_norm` reaches 0.6460. The paired gap is $+38.76$ points with McNemar $p = 9.8\times10^{-148}$. The arm can rank answer contents but cannot express that knowledge through the damaged letter interface. Conversely, healthy models often favor letters, so "content is the fair interface" is not a general conclusion.

The residual-fraction effect is also construct-specific. On OpenBookQA `content_norm`, using chance rather than the 0.3680 token-longest floor inflates the base arm's residual fraction by 2.11×. PIQA and ARC-Easy move in the opposite direction, to 0.90× and 0.98×, because their token-longest floors are below chance. Null calibration can increase or decrease the residual; it is not a one-way correction.

5.3 V2 RE-SORTS THE 27-CELL READ-OUT

Under v1, two MMLU-Pro cells are significantly below the best-constant floor. Under v2, the below-null count is zero. `qwen3/k8` moves from -0.881 points ($p = 0.0362$) to -0.139 ($p = 0.0964$), and `llama2/k8` from -0.914 to -0.416 ($p = 0.1002$). Their v1 verdicts remain valid statements about an arm-independent interface floor, but they are not evidence that either arm is worse than its own input-blind prediction marginal. Both competence claims dissolve.

V2 is not uniformly conservative (Table 6, Appendix A.8). It withdraws the published above-floor capability label for `qwen3/k14`, replacing it with `TRACE_SIGNAL`, while `olmo2/keep14` moves from at-floor to `TRACE_SIGNAL`. The two moves are not equally defensible and we separate them rather than reading them as one bidirectional effect. Table 7 carries a Benjamini–Hochberg q-value over exactly its own 27 cells, which is the one place in this paper where the family definition is not in dispute: one table, one null, one statistic, one item set. The downgrade survives correction — `qwen3/k14` has $q_{\mathrm{BH}} = 0.0297$ — but the upgrade does not: `olmo2/keep14` at $p = 0.0172$ sits at rank 7, where the step-up threshold is $7 \times 0.05/27 = 0.0130$, giving $q_{\mathrm{BH}} = 0.0663$. Under Bonferroni ($\alpha/27 = 0.001852$) neither trace cell clears and only the five $p = 0.0001$ anchors do. Since `olmo2/keep14` is the *only* cell moving upward, the corrected read-out re-sorts in one direction rather than two. We therefore claim only what survives: v2 does not merely rescale v1, because it reverses one verdict that holds up under multiplicity correction over its own family. The upward move is reported as a single uncorrected per-cell observation and no conclusion in this paper rests on it.

It also exposes a benchmark-level limitation invisible to v1. Intact Llama-2-7B has `recovery_fraction`=0.0545 on MMLU-Pro, even though v1 calls it comfortably above floor by $+1.538$ points. Because the intact anchor itself is below 0.10, relative-recovery claims for the Llama-2 family are blocked.

Significance alone would re-import the defect. At $n = 12032$, `qwen3/k14` reaches $p = 0.0066$ at only $+0.267$ points while A wins the likelihood argmax on 94.6% of items. The same data reject simpler diagnostics (Table 8): the A01 own-modal null over-credits P1 by 1.37 points and flips its sign; for Qwen3 heal@7000 and unhealed `k8`, content is -3.610 and -1.076 points under the same permutation null, compared with letter at -0.022 and -0.139; and `llama3/k12` has modal share 0.339 and normalized entropy 0.614 yet $\Delta_{\mathrm{perm}} = +0.002$ points ($p = 0.997$). Entropy and modal share are descriptive, not competence metrics.

5.4 FULL PRECISION FALSIFIES THE NUMERICAL-TIE MECHANISM

For OLMo-2 `keep8` on MMLU, fp32 removes all 4,303 bf16 exact top-two ties and changes 2,532 of 14,042 letter argmax decisions, or 18.03%. Letter accuracy nevertheless changes by only -0.0015, CI95 $[-0.0064, +0.0033]$, exact McNemar $p = 0.5702$. The arm is more significantly below its floor in fp32: -1.538 points, $p = 0.0060$, versus -1.389, $p = 0.0190$, in bf16. Higher precision can reshuffle ambiguous decisions; it cannot create item-level information that is absent.

Auditing the audit. The analysis uncovered and repaired three defects without changing the substantive verdicts: a doubled-tail bootstrap formula that produced an illegal $p = 1.042$ (re-emission changed 0/24 verdicts and moved 0/30 p-values across 0.05), a sequence cap validated on the wrong tokenizer that silently left-truncated 10/15 MMLU-Pro cells (0/14 verdicts changed, at most 0.0083 points, but benignity was unknowable in advance), and an out-of-memory failure on 5/8 intact-Llama-2 shards that the merge guard correctly refused. All reported cross-family MMLU-Pro results use the corrected launch. Table 9 and Appendix A.7 give each defect with its measured impact.

6 Discussion

A reporting rule, not a universal theory of failure. The practical recommendation is small: define the strongest construct-appropriate constant or input-blind predictor, report its value and specification, test the arm against it, and only then compare arms. The rule does not say every damaged model becomes constant, every MC interface is invalid, or every chance line inflates a claim. It says that an arm score without its appropriate floor can support the wrong verdict.

The letter-floor and permutation read-outs partition scope. V1 must be arm-independent because it certifies mutual comparability of an interface. V2 must be arm-conditional because it asks whether one prediction vector aligns with its items beyond its own output marginal. Substituting one for the other creates the defects we observe: chance credits constants, while a best-constant competence test confounds knowledge with collapse-letter identity.

Relationship to prediction debiasing. PriDe-style option permutations estimate and remove a model's selection bias (Zheng et al., 2024). This is complementary, not redundant. A corrected predictor can still be evaluated against the wrong reference; an explicit floor remains necessary. Conversely, a floor test does not repair a biased predictor. A complete evaluation can report both predictor-side debiasing and reference-side null calibration.

Power is part of construct validity. Small benchmarks produce apparently reassuring at-floor results even when observed effects are large. Any interpretation of a null result must include the achieved interval. MMLU-Pro changes the evidential status: OLMo-2 `keep8`'s interval excludes an MMLU-sized below-floor effect, so this is a powered non-replication, not an absence of evidence. The broader ladder remains a cliff rather than a gradient; no depth curve should be fitted to the sparse rungs.

Current claim versus retracted claims. The flat ledger in Appendix A.13 (Table 11) records the self-falsification history. The surviving claim is narrower and stronger: under structural damage, the letter interface frequently fails its own arm-independent best-constant floor while remaining above a conventional chance line. This is 15/17 rather than universal at MMLU-Pro, is not equivalent to literal constant emission, and does not by itself identify healing or family as a cause. Its exceptions are informative rather than tolerated: off MMLU every cell clearing its floor is a healed arm retaining at least 14 of 32 layers, so the failure tracks damage severity.

A healed non-OLMo arm is in training, but it is not part of this paper's evidence. Its remaining question is only whether it develops material item-level signal. The original `H_heal`/`H_family` dichotomy is unavailable because its unhealed comparator does not fall below the collapse-proof permutation null.

7 Limitations

Character-unit reconstruction. The OpenBookQA character floor was added after the original audit identified an evidence hole. It is now recomputed from raw parquet and passes a 12/12 character/token self-test, so we do not use the earlier "unrecomputable" status. The remaining limitation is narrower: the character values are reconstructed from raw option text rather than retained as per-item fields in the scored records. Item matching to the token leg therefore follows the shared loader and self-test rather than a stored character-count column. The convention of excluding the continuation's leading space was explicit; including it yielded the identical 0.363500 value.

One family at one depth cannot identify healing. The prospective healed contrast contains one non-OLMo family at one absolute depth. Llama-2, Llama-3, and the `k10`/`k12`/`k14` rungs remain confounded. The observed ladder is a cliff, not a gradient, and we fit no depth curve.

The healing corpora are unmatched and cannot be matched from available data. The Qwen3 arm heals on 5.72 epochs of 5.541B SlimPajama tokens, whereas OLMo-2 used 1.0 epoch of 31.7B Dolmino tokens. Qwen3 cannot consume OLMo-2-token Dolmino and raw Dolmino text is on neither disk. Relative depth is also untested: Qwen3 has 36 layers and OLMo-2 has 32, so `keep8` retains 22.2% versus 25.0%.

The planned causal contrast has dissolved. `H_heal` is not merely unmeasured. Its antecedent required the unhealed Qwen3 `k8` comparator to sit below an arm-conditional null; it does not (-0.139 points, $p = 0.0964$). No amount of further training repairs a comparator that fails the criterion. `H_family` is equally unavailable because 0/27 cells are significantly below the permutation null. A future read-out can ask only whether the healed arm has material item-level signal.

Regime, pooling, and literature scope. Non-OLMo cross-family arms are truncate-only, whereas OLMo-2 arms are prune-then-heal; tables that include both cannot identify a family effect. Pooled results average per-item deviations over disjoint benchmarks and must not be read as per-benchmark verdicts. Finally, the Cho et al. overlap audit used arXiv v4 dated January 12, 2026; the January 26, 2026 camera-ready could not be diffed because the OpenReview PDF endpoint was inaccessible from the audit network.

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8 Ethics Statement

This work studies evaluation methodology and uses existing benchmark records and base-model checkpoints. It introduces no human-subject data collection and makes no deployment claim about the evaluated models. The main ethical risk is epistemic: a poorly calibrated reference can make a damaged or degenerate system appear capable, which may distort model comparisons and downstream decisions. We mitigate that risk by publishing the null definition, uncertainty, power, integrity checks, counterexamples, and a flat retraction ledger. We also avoid treating a floor pass as certification, since untested artifacts may remain.

The experiments consumed substantial accelerator resources in their original execution. The manuscript itself adds no new training or scoring result. Our recommended nulls are primarily computed from existing labels, option metadata, and prediction records, so they can usually be added with negligible compute relative to model evaluation.

Reproducibility Statement

Every quantitative claim in this paper is bound to a machine-readable record, and the binding is checked mechanically rather than by hand. A checker walks all numerals in the source and requires each to be either an exact match to a value in the evidence set, a correct rounding of one, or a stated difference of two; at the time of writing it reports 610 numerals with none unresolved. The two nulls, the winner's-curse calibration of the best-constant floor, the stratified permutation statistic, and the power analysis are each emitted from the per-item prediction records by a script rather than transcribed, so a reader who re-runs the emitters obtains the tables as printed or a diff.

The measurement protocol is fixed and reported in full in Section 4: `chat_template = False`, `add_bos = 0`, `desc_style = none`, fp32 master weights with bf16-autocast forward, batch size 48, and maximum length 2048 for the cross-family MMLU-Pro runs, with zero truncation asserted per shard in every reported cell. The chat template is disabled deliberately and not for convenience: the evaluated systems are base language models with neither supervised fine-tuning nor RLHF, so applying a chat template would measure a different object.

Three properties of this study are load-bearing for reproduction and are stated here rather than left implicit. First, the floor is a maximum over noisy label marginals and is therefore upward biased even under exactly uniform labels; any attempt to reproduce a floor-above-chance claim must reproduce that calibration, not only the point estimate. Second, content-side floors are not a property of a dataset alone: they are determined jointly by the tie convention, the length unit, and the tokenizer, so a reproduction that fixes only the dataset will not recover our numbers. Third, the ordering between the two nulls is a measured property of the reported cells under an explicitly stated regularity condition, not an identity; we give the margin by which it holds and the construction that violates it.

We also record what a reproduction will *not* find. Claims that earlier drafts of this work considered and that the evidence did not sustain are listed with their current status in Table 11, including six retracted and two prohibited numeric claims. That ledger is part of the artifact: it exists so that a reader who encounters one of those numbers elsewhere can see that we withdrew it and why.

A Appendix

A.1 Four Under-Specifications of the Reference

This appendix develops the four degrees of freedom summarised in Section 3.3, one subsection each, and carries the two tables they are read from. The phrase "compare to a baseline" hides choices large enough to reverse a verdict. These choices are part of the measured construct and must be printed with the score (Table 2).

A.1.1 Tie convention

For a longest-option heuristic, tied maxima can be split fractionally, resolved by the first or last option, credited whenever gold belongs to the tied set, or counted as wrong. On ARC-Challenge, `split`, `first`, `last`, and `wrong` place all six OLMo-2 arms significantly above the content floor; `credit` moves five of six significantly below it. On MMLU-Pro, the same dataset, tokenizer, and length unit yield 0.125914 under `wrong` and 0.532164 under `credit`, a 40.6-point span. The `credit` value is 4.2264× the `wrong` value and exceeds the intact base model's `content_norm` score, 0.207613, by 32.5 points. A convention that lets an oracle harvest every tie is useful as a reductio, not as a neutral default.

A.1.2 Length unit

"Longest" may mean characters or continuation tokens. On identical items, `split` changes by as much as 2.02 points across the studied tasks, while `credit` changes by 14.53–35.20 points. For OpenBookQA, `credit` is 0.416 under characters and 0.644 under tokens. The reason is visible in the tie sets: ARC-Challenge has 18.60% tied-longest items under characters and 50.85% under tokens. Thus printing the convention without the unit is not reproducible specification.

The character-unit values were recomputed from raw parquet under $\mathrm{len}(\text{option text})$. Counting the continuation's leading space gives the same OpenBookQA value, 0.363500, because the shift is uniform across options. The character and token legs are item-matched by the shared loader and a 12/12 self-test.

A.1.3 Tokenizer

Within the token unit, the floor is tokenizer-valued. On identical items, the `split` floor moves by 0.9003 points in the audited comparison and `credit` moves by up to 10.6 points on four-way tasks and 9.26 points on MMLU-Pro. This is a floor-level consequence of the known tokenizer dependence of length-based scoring (Oostermeijer, 2026). We make no vocabulary-size explanation: the ordering is not monotone. The 32k Llama-2 and 152k Qwen3 tokenizers produce fewer MMLU-Pro longest-option ties than the 100k OLMo-2 and 128k Llama-3 tokenizers.

This degree of freedom creates an asymmetry. A token-based content floor is a property of $(\text{dataset}, \text{convention}, \text{unit}, \text{tokenizer})$. A character-based floor is tokenizer-free under its stated character convention. By contrast, the letter floor is a pure property of the item set's gold labels: it is invariant across all 15 cross-family arms and bit-identical across all 21 MMLU-Pro cells.

A.1.4 What "chance" means when option count varies

MMLU-Pro is described as up to ten-way, but n_opt ranges from three to ten. Naive $1/10$ chance is 0.100000, whereas the item average $\mathrm{mean}(1/\texttt{n\_opt})$ is 0.110877. Against the same always-A floor, 0.116606, the gap is therefore either $+1.661$ points and $1.1661\times$, or $+0.573$ points and $1.0517\times$. Both are defensible descriptions of a chance line, so both must be reported. The floor remains a dataset property; its gap to an under-specified chance line does not.

A.1.5 The calibrating null must be realisable, and we initially failed this on our own largest construct

The four degrees of freedom above concern the *floor*. A fifth concerns the null used to calibrate it, and we get it wrong in the obvious place. Our first calibration drew all 12032 MMLU-Pro gold letters uniformly over all ten letters A–J. But n_opt varies, so 2051 of 12032 items (17.05%) have fewer than ten options and cannot have gold letter J at all. That null therefore places its mass *outside the support of every legal labelling of the benchmark*: it is not a conservative approximation of the right experiment but a different experiment. The consequence is not cosmetic. Under a uniform-over-ten null the ten letter marginals are exchangeable; under any legal labelling the low letters, legal on every item, carry systematically more mass than the high letters, legal only on the wide items ($\mathbb{E}[\hat{m}_A] = 0.110877$ against $\mathbb{E}[\hat{m}_J] = 0.082954$). Exchangeability understates $\mathbb{E}[\max_L \hat{m}_L]$, and it is precisely that maximum which the floor must be compared against.

The admissible null holds the observed n_opt histogram fixed — n_opt is a property of the item set, not a random variable — and draws each item's gold letter uniformly among *its own* legal letters. On MMLU-Pro this raises $\mathbb{E}[\hat{f}]$ from 0.104457 to 0.113873, a 0.94-point understatement, and moves the row from $p < 10^{-5}$ to $p = 0.083$. The observed floor does not move at all: it is $\max_L(\mathrm{count}_L/n) = 1403/12032$, computed from the gold labels with no null involved. So the correction changes what the floor is compared *to*, never the floor itself, and every downstream count that is defined against the observed floor — the 15/17 aggregate, the 3/12-versus-1/12 symmetric-standard flip, and the off-MMLU 9/85 — is untouched by it.

Which rows moved, and in which direction. We state this explicitly because a correction that a reader cannot audit for self-service is not a correction. Where n_opt is constant the two nulls are the *same distribution*, so no correction is possible and none is applied: MMLU is exactly four-way on all 14042 items and BoolQ exactly two-way on all 3270, and both retain $p < 10^{-5}$; OpenBookQA, CommonsenseQA and PIQA are likewise fixed at four, five and two. The two surviving constructs are therefore exactly the two whose option count was never in question, which is the opposite of a convenient outcome: the row we lose is the largest construct we have and the one that carries the power analysis. Where n_opt varies the sign depends on which way it varies, and both directions occur here. Items with *fewer* options than nominal concentrate mass on the always-legal low letters, raising $\mathbb{E}[\hat{f}]$ and hence p; this is MMLU-Pro, overwhelmingly, and it moves against us. Items with *more* options than nominal divert mass to a letter that is legal almost nowhere else and so can never attain the maximum, which lowers $\mathbb{E}[\hat{f}]$ and hence p; this is ARC-Easy, where 4 of 2376 items are five-way, and ARC-Challenge, where 3 of 1172 are. Those two rows move slightly *in our favour*, by -0.0025 and -0.0063 in p — real rather than Monte-Carlo error at roughly -17 and -23 standard errors, but far too small to approach the conventional threshold from either side ($0.140 \rightarrow 0.136$ and $0.453 \rightarrow 0.447$), so no verdict turns on them. We report them rather than absorbing them because the general claim that such a fix can only ever move against the authors is false, and we made it before measuring.

A rounding convention with the same sign problem. Since $p = \Pr(\hat{f} \geq \text{floor})$ and the floor is an exact rational count/n, the threshold is an integer count. Storing the floor rounded to six decimals and comparing against the rounded value can demand one *extra* count: $0.116606 > 1403/12032$, so the stored form excludes the very outcome that was observed, biasing p downward — again toward "survives". We therefore evaluate every p at the exact rational floor. On MMLU-Pro this gives 0.083 rather than 0.078 and on PIQA 0.691 rather than 0.658. The hazard covers five of the table's nine rows — four distinct constructs, since MMLU-Pro contributes a row under each chance convention — and not the two whose p moved: MMLU's stored 0.268908 likewise exceeds $3776/14042$, and BoolQ's 0.621713 exceeds $2033/3270$, so on those the rounded threshold also demands a count that

Construct	n	$L^\star$	Floor	Chance	Gap (pp)	Ratio	$\mathbb{E}[\hat{f}]$	q_{95}	p
MMLU-Pro, naive	12032	A	0.116606	0.100000	+1.661	1.1661×	0.113873	0.117188	0.083
MMLU-Pro, item-avg.	12032	A	0.116606	0.110877	+0.573	1.0517×	0.113873	0.117188	0.083
MMLU	14042	D	0.268908	0.250000	+1.891	1.0756×	0.254350	0.258225	$< 10^{-5}$
OpenBookQA	500	A	0.276000	0.250000	+2.600	1.1040×	0.273190	0.294000	0.384
ARC-Easy	2376	C	0.266414	0.250161	+1.625	1.0650×	0.260523	0.269781	0.136
ARC-Challenge	1172	B	0.265358	0.250156	+1.520	1.0608×	0.264984	0.278157	0.448
CommonsenseQA	1221	B	0.208845	0.200000	+0.884	1.0442×	0.214994	0.226863	0.853
PIQA	1838	B	0.504897	0.500000	+0.490	1.0098×	0.509300	0.522851	0.691
BoolQ	3270	B	0.621713	0.500000	+12.17	1.2434×	0.506972	0.517125	$< 10^{-5}$

Table 1: Construct floors compared with chance, *and with the sampling distribution of the floor estimator itself*. Both percentage-point gaps and ratios are needed across option counts. The last three columns calibrate the estimator. Because $\hat{f} = \max_L \hat{m}_L$ is a maximum over k noisy marginals estimated on a single finite item set, it is upward biased even when the gold labels are exactly uniform, so a positive Gap is not by itself evidence that the floor differs from chance. $\mathbb{E}[\hat{f}]$ and q_{95} are the mean and 95th percentile of $\hat{f}$ under a balanced null *that the construct can actually realise*: the observed `n_opt` histogram is held fixed and each item's gold letter is drawn uniformly among *its own* legal letters, so an item with four options is never assigned gold letter J. Where `n_opt` is constant this coincides exactly with drawing uniformly over all k letters; where it varies — both MMLU-Pro rows, ARC-Easy and ARC-Challenge — it does not, and the uniform-over-k version we reported previously lies outside the support of any legal labelling (Appendix 10 keeps that superseded column beside this one, and Section A.1.5 states what it changed and in which direction). $p = \Pr(\hat{f} \geq \text{observed floor})$ under the realisable null, evaluated at the floor's exact rational value count$/n$ (10^6 draws per construct, three seeds, two independent implementations; the reported $p < 10^{-5}$ bound is the more conservative one carried over from the earlier 2×10^5-draw run; evidence E-CAL, the single machine-readable source for these rows). **Only MMLU and BoolQ have floors a balanced null could not plausibly produce** ($p < 10^{-5}$); both are constructs with a genuinely fixed option count, so for them the choice of null is not in question. For the six remaining letter constructs the observed floor lies inside the estimator's own noise ($p = 0.083$–0.853), and on CommonsenseQA and PIQA it is in fact *below* $\mathbb{E}[\hat{f}]$. Those six rows therefore support the weaker claim that chance is the wrong *reference*, while providing no evidence that the floor *differs from* chance in magnitude; the $+0.49$–$+2.60$ pp gaps there are the regime our own evidence record marks "do not oversell" (evidence E-A). MMLU-Pro is the closest of the six to the conventional threshold and should be read as unresolved at this n rather than as settled either way. Floors and chance values are the label-marginal floors re-derived first-hand from the raw per-item records (evidence E-A; see the reproducibility statement for the artifact paths).

was never observed. Their verdicts are untouched only because both sit at $p < 10^{-5}$, where a one-count shift cannot reach the conclusion — a fact about how far those rows are from the boundary, not evidence that the convention is harmless where it applies. The remaining rows are genuinely clean: OpenBookQA's floor is exact at $138/500$, and ARC-Easy, ARC-Challenge and CommonsenseQA round down. No verdict changes, but MMLU-Pro's distance from 0.05 does, and it moves away from significance rather than toward it.

What this leaves standing. The protocol claim is unchanged and arguably strengthened: a null must be compatible with the construct it calibrates, and the failure mode is available to any author who writes "uniform over k" for a benchmark whose k is not constant. What we withdraw is the quantitative claim for one construct. We no longer assert that MMLU-Pro's floor exceeds what a balanced null could produce; at $p = 0.083$ it is unresolved at $n = 12032$, close enough to the conventional line that a larger item set could plausibly settle it, and we decline to describe it as either significant or comfortably inside the noise.

Dataset	Unit/tokenizer	`split`	`first`	`last`	`credit`	`wrong`
MMLU-Pro	token, OLMo-2	0.193150	0.195894	0.190824	**0.532164**	0.125914
OpenBookQA	character	0.363500	0.362000	0.362000	0.416000	0.320000
OpenBookQA	token, OLMo-2	0.368000	0.384000	0.360000	**0.644000**	0.228000
ARC-Challenge	character	0.274104	0.272184	0.274744	0.347270	0.220990
ARC-Challenge	token, OLMo-2	0.283902	0.265358	0.296075	**0.543515**	0.142491
Winogrande, control	character	0.491713	0.492502	0.490923	0.581689	0.401736
Winogrande, control	token, OLMo-2	0.501184	0.494081	0.508287	**0.933702**	0.068666

Table 2: The content floor changes with tie convention and length unit. Numbers are the content-side longest-option floors recomputed per (convention, unit, tokenizer) from the per-item option records, including the raw-parquet character reconstruction and its character/token self-test (evidence E-F, E-H and the August 14 character-unit addendum; see the reproducibility statement for the artifact paths).

A.2 Prior art on option-count-aware chance correction

Section 3.2 claims the within-stratum permutation and its use as a gate, and nothing else. This appendix attributes the rest, result by result. Bennett's S sets $p_e = 1/k$ for k response categories (Bennett et al., 1954); Brennan & Prediger (1981) recommend exactly that free-marginal alternative wherever κ's marginal dependence misleads; formula scoring corrects for guessing as an explicit function of the per-item option count (Frary, 1988); and Brenner & Kliebsch (1996) show that a weighted κ drifts systematically with the number of categories — the very pathology our stratification is designed to avoid. Pooling an agreement coefficient across heterogeneous item groups is likewise established (Vries et al., 2008). We therefore claim *none* of the following: correcting for the number of options, choosing p_e as a methodological decision, or noticing that κ-family statistics depend on k. The works above assume a single global k; what we are not aware of in that literature is a null in which *k varies item to item* ($\texttt{n\_opt} \in \{3, \ldots, 10\}$, eight strata here), with the permutation confined within strata so that a letter illegal for an item can never be credited to it. That is a loophole-closing construction, not a new statistic, and its measured footprint on MMLU-Pro is the 36 items quantified in Appendix A.3.

A.3 Why the v1/v2 ordering is not an identity

Section 3.2 states the ordering between the two nulls as a measured property under an explicit regularity condition. This appendix gives the argument in full, and Table 3 sets the two read-outs side by side.

Unstratified, $\sum_L p_L m_L \leq \max_L m_L$ places the v2 null at or below the v1 best-constant floor. Our estimator, however, permutes *within* $\texttt{n\_opt}$ strata, and stratification only yields $\widehat{\text{acc}} \leq \sum_s w_s \max_L m_{s,L}$, whose right-hand side dominates $f_{\text{const}} = \max_L \sum_s w_s m_{s,L}$. The v1 ordering is therefore recovered only if $\arg\max_L m_{s,L}$ is the same letter in every stratum, which on MMLU-Pro it is not: the global best constant is always-A ($1403/12032$), but the per-stratum argmax is B, E, B, E for $\texttt{n\_opt} = 5, 6, 7, 8$ (4 of 8 strata, 623 of 12032 items), giving $\sum_s w_s \max_L m_{s,L} = 1439/12032$ and a gap of exactly 36 items, or 0.299 pp. Empirically the ordering holds in all 27 cells of Table 7, but that count overstates the evidence: 13 of the 27 clear the floor by more than 0.299 pp and so could not have violated it under *any* prediction vector, while the remaining 14 come from four base models, five of them one healing run sampled 500 steps apart. The binding comparison is per-cell: for the tightest cell (Llama-2 keep-12, un-healed, margin 0.0078 pp) the largest violation attainable at its own observed letter marginal is 0.085 pp, because 94.8% of its weight lies in A-argmax strata. The ordering does become a theorem under a condition we can state but not assume — that the arm's letter distribution not vary with $\texttt{n\_opt}$, since $p_{s,L} = p_L$ gives $\widehat{\text{acc}} = \sum_L p_L f_L \leq f_{\text{const}}$ exactly — so these cells are evidence for *that condition*, not for the bound. Nor is the inversion confined to other benchmarks: on MMLU-Pro itself the $\texttt{n\_opt}$-conditional emitter A,A,B,E,B,E,A,A is legal on every item and attains $f_{\text{const}} + 0.299$ pp. We therefore state the ordering as a measured property of these cells under an explicit regularity condition, not as an identity.

Read-out	Question	Null	Required dependence
v1, headline	Is this interface valid for comparing arms?	Best constant, e.g. always-A 0.116606	Arm-independent
v2, companion	Does this arm's prediction vector contain item-level information?	Prediction vector permuted within `n_opt` strata	Arm-conditional

Table 3: Two nulls answer two different questions. Values and definitions are taken from the stratified permutation-null record for the re-judged cells (evidence E-D; see the reproducibility statement for the artifact path).

A.4 Designated damaged cells, multiplicity, and power

Designated damaged cells. Throughout, *designated damaged* means an arm that was structurally pruned and is reported as a damaged rung of its family; the designation is fixed by the arm's construction, not by its measured score. The set is stated once and used everywhere: the five OLMo-2 prune-then-heal arms `shortgpt16`, `keep14`, `keep12`, `keep10`, `keep8`, and the four truncation rungs `k14` through `k8` of each of Llama-2-7B, Llama-3-8B and Qwen3-8B-Base, giving 17 MMLU-Pro cells and $17 \times 5 = 85$ off-MMLU cells; Winogrande is the sole negative control and enters no denominator. Where a denominator such as 15/17 or 9/85 is quoted, it is over exactly this set, and any damaged arm excluded from that denominator is named at the point of exclusion together with its own floor delta, so that no ratio is computed over an undisclosed subset. An earlier version of this paper violated that rule: the MMLU-Pro roll-up counted only `keep12`, `keep10` and `keep8` as the OLMo-2 damaged set, so the quoted 14/15 and 0/15 silently omitted `shortgpt16` (+3.674 points above its floor, $p = 0.0001$) and `keep14` (+0.324, $p = 0.3234$). The omission was not defensible as a rule, because the identical retained depth of 14 layers *was* counted as damaged for all three non-OLMo families inside the same ratio. Both arms are now in every denominator, and `gate_designated_denominator.py` asserts that the arm set declared in Section 4 equals the set each roll-up counts, that no retained depth is damaged in one family and intact in another, and that every denominator's cardinality equals arms times benchmarks.

Multiplicity. We report per-cell $\alpha = 0.05$ decisions across many cells without a family-wise correction, and the headline counts are therefore counts of per-cell decisions rather than simultaneously valid claims. This is a deliberate and stated limitation: the cells share items, nest arms, and share a null, so the tests are neither independent nor exchangeable, and an off-the-shelf Bonferroni or Benjamini–Hochberg adjustment would not have a defensible family definition here. Readers should treat any individual borderline cell as a per-cell decision. The paper's conclusions rest on the direction of the aggregate and on the regime-and-depth pattern that organises its exceptions — 9/85 off-MMLU cells clear their floor and all nine are healed arms retaining at least 14 of 32 layers — rather than on any single cell crossing α. We note that this is a weaker substitute for multiplicity control than an exceptionless aggregate would have been, and that the exceptions are concentrated rather than scattered is itself the load-bearing observation: a test that failed indiscriminately would not place all of its rejections at one end of the damage ladder.

Table 4 is the power disclosure that Section 4 requires before any null result on the five smaller benchmarks is interpreted.

Task	n	`keep8` Δ	CI95 half-width	Detect -1.389 pp?
MMLU	14042	-1.389	1.154	reference
ARC-Easy	2376	-0.800	1.305	yes, borderline
Winogrande, control	1267	$+0.552$	1.184	yes
PIQA	1838	$+2.503$	2.775	no
CommonsenseQA	1221	-1.065	3.399	no
ARC-Challenge	1172	-0.939	3.882	no
OpenBookQA	500	-1.800	6.400	no

Table 4: Mandatory power disclosure for small-benchmark null results. Numbers are the paired-bootstrap half-widths recorded per cell in the five-benchmark letter/content null records (evidence E-F; see the reproducibility statement for the artifact path).

A.5 Multiplicity in the two-reference comparison

Two qualifications belong with the two-reference comparison of Section 5.1. First, one instrument does not transfer: the floor comparison also carries an exact McNemar test, which has no analogue here because $\mathrm{mean}(1/\mathtt{n\_opt})$ is not a $0/1$ predictor and so generates no discordant pairs. Second, these remain per-cell decisions in the sense of Appendix A.4, and here the qualification bites harder than it does for the near-unanimous aggregates: with 12 tests at $\alpha = 0.05$ the expected false-positive yield is 0.60, and under Benjamini–Hochberg at $q = 0.05$ — or Bonferroni — *neither* reference retains a single cell (0/12 both sides), so the corrected comparison is undefined rather than 3/12 versus 1/12. What survives correction is the count itself: observing 3 or more rejections out of 12 has binomial probability 0.0196 under the global null. We therefore read this comparison as evidence that the chance side is not uniformly null, not as a set of three simultaneously valid per-cell claims.

A.6 Constant emitters and the off-MMLU replication

The most vivid small-benchmark cases are literal constant emitters. Sixteen damaged cross-family cells have accuracy equal to the marginal of the emitted letter to machine precision. Two OpenBookQA cells emit A on every item, score exactly 0.276000, have $\Delta = 0.000$ points and CI95 $[0, 0]$, and therefore land exactly on the optimal constant. Against chance 0.25, the same empty behavior appears 2.6 points above chance.

Off MMLU, the full designated set of 85 cells gives 45/85 above chance and 9/85 above floor. The nine are not scattered: all of them are OLMo-2 arms retaining at least 14 of 32 layers, five from `shortgpt16` ($+11.21$ to $+49.66$ points, every cell at $p = 0.0001$) and four from `keep14` ($+6.47$ to $+18.69$, $p \leq 0.0005$; its fifth cell, PIQA at $+2.61$, is at the floor). Restricting to OLMo-2 arms that retain 12 layers or fewer gives 0/15, and the three truncation families give 0/60 at every depth they cover, including the 0/15 at retained depth 14 that matches `keep14`'s depth without matching its behaviour — so the threshold is a property of the healed ladder, not of depth alone. The two groups do not overlap: the smallest above-floor margin among the nine is $+6.47$ points while the largest point estimate among the 76 at-or-below cells is $+2.61$. Within the non-OLMo replication, 25/60 read above chance and only 7/60 are significantly below because 52/60 are underpowered for the MMLU reference effect. This is not an effect-size explanation: ARC-Challenge's median damaged effect is -3.840 points, larger than MMLU's -3.603, but its half-width is 3.92 rather than 1.18 points. Regime and family remain confounded here as everywhere in this paper: OLMo-2 is the only prune-then-heal family, and no truncation rung retains more than 14 layers, so the depth threshold is a property of the OLMo-2 ladder and must not be read as a family contrast.

A.7 Auditing the audit: the three defects in full

The analysis uncovered and repaired defects without changing the substantive verdicts; Table 9 collects them with their measured impact. A doubled-tail bootstrap formula counted zero-valued resamples in both tails and produced an illegal $p = 1.042$. Splitting the zero atom evenly makes $p \leq 1$ structural; re-emitting the summaries changed 0/24 verdicts, moved 0/30 p-values across 0.05, and left every non-p field byte-identical.

Family	Arm/regime	Accuracy	Δ vs floor (pp)	CI95 half-width	p	v1 verdict
OLMo-2	`shortgpt16`, heal	0.153341	$+3.674$	0.702	0.0001	above floor
OLMo-2	`keep14`, heal	0.119847	$+0.324$	0.636	0.3234	at floor
OLMo-2	`keep12`, heal	0.113115	-0.349	0.765	0.3805	at floor
OLMo-2	`keep10`, heal	0.112450	-0.416	0.727	0.2662	at floor
OLMo-2	`keep8`, heal	0.115442	-0.116	0.582	0.7118	at floor
Llama-2	`k14`, truncation	0.116772	$+0.017$	0.083	0.7072	at floor
Llama-2	`k12`, truncation	0.115858	-0.075	0.083	0.0693	at floor
Llama-2	`k10`, truncation	0.113198	-0.341	0.416	0.1132	at floor
Llama-2	`k8`, truncation	0.107463	-0.914	0.736	0.0168	below floor
Llama-3	`k14--k8`, truncation	0.111868–0.114694	-0.474–-0.191	0.407–0.852	0.1296–0.6631	all at floor
Qwen3	`k14`, truncation	0.118933	$+0.233$	0.191	0.0192	above floor
Qwen3	`k12`, truncation	0.115027	-0.158	0.428	0.4645	at floor
Qwen3	`k10`, truncation	0.118185	$+0.158$	0.233	0.1872	at floor
Qwen3	`k8`, truncation	0.107796	-0.881	0.819	0.0362	below floor

Table 5: MMLU-Pro letter read-out against the bit-identical always-A floor 0.116606, over all 17 designated damaged cells. OLMo-2 arms are prune-then-heal; the other families are truncate-only, so regime and family are confounded. Rows are ordered by retained depth within family: the two cells that clear the floor are the OLMo-2 arm retaining 16 of 32 layers and Qwen3 `k14`, and the remaining 15 are at or below it. Numbers are the per-cell MMLU-scale power records for the letter-floor cells (evidence E-B; see the reproducibility statement for the artifact path).

The first cross-family MMLU-Pro launch also had two integrity failures. A sequence cap validated on the OLMo-2 tokenizer silently left-truncated 10/15 cells; raising it to 2048 and making `n_trunc` a hard per-shard assertion changed 0/14 verdicts and at most 0.0083 points, but benignity was unknowable in advance. Separately, intact Llama-2 exhausted memory on 5/8 shards because it lacks grouped-query attention; the merge guard correctly refused the 3/8 partial result. Setting `use_cache=False` recovered the cell. All reported cross-family MMLU-Pro results use the corrected launch.

A.8 Per-cell MMLU-Pro read-out and the v2 re-sort

The two per-cell tables cited from the Results and Analysis section are placed here for space. Table 5 is the MMLU-Pro letter read-out for all 12 cross-family cells against the always-A floor, supporting "Wrong-null verdict flips at MMLU-scale power". Table 6 gives the paired v1/v2 verdicts for the five cells whose labels the arm-conditional null moves, supporting "V2 re-sorts the 27-cell read-out"; Table 7 gives all 27 cells with Benjamini–Hochberg q-values over exactly that family, which is what separates the surviving downgrade from the non-surviving upgrade. Every value in these tables is unchanged from the version those subsections discuss.

Cell	v1 Δ	v1 p	v1 verdict	Δ_{perm}	v2 p	v2 verdict
Qwen3 `k8`, unhealed	-0.881	0.0362	below floor	-0.139	0.0964	no item-level signal
Llama-2 `k8`, unhealed	-0.914	0.0168	below floor	-0.416	0.1002	no item-level signal
Qwen3 `k14`, unhealed	$+0.233$	0.0192	above floor	$+0.267$	0.0066	trace signal
OLMo-2 `keep14`, healed	$+0.324$	0.3234	at floor	$+0.608$	0.0172	trace signal
Llama-2 intact	$+1.538$	0.0001	above floor	$+1.959$	0.0001	anchor blocked, recovery 0.0545

Table 6: V2 re-sorts rather than uniformly deflates the read-out. All deltas are percentage points. Numbers are the paired v1-floor and v2-permutation records for the same cells (evidence E-D; see the reproducibility statement for the artifact path).

A.9 Complete read-out v2 table

Table 7 reports all 27 cells used in the post-hoc v2 re-analysis. G1 and G2 fire on no cell; all 27 pass the integrity gate; the permutation and bootstrap decisions disagree on 0/27 cells at $\alpha = 0.05$. The largest absolute difference between their p-values is 0.0371, away from the decision boundary. Stratification is nevertheless load-bearing: for unhealed Qwen3 `k8`, the stratified statistic differs from the unstratified one by -1.271 points because E is not legal on every item.

Family	Cell	Acc.	$\widehat{\text{acc}}$	Δ_{perm}	Half-width	p	q_{BH}	Verdict
Qwen3	heal@5000	0.115276	0.115993	-0.072	0.231	0.5412	0.7306	no signal
Qwen3	heal@5500	0.115691	0.115654	$+0.004$	0.316	0.9810	0.9968	no signal
Qwen3	heal@6000	0.114860	0.115857	-0.100	0.281	0.4932	0.7306	no signal
Qwen3	heal@6500	0.114943	0.115864	-0.092	0.287	0.5204	0.7306	no signal
Qwen3	heal@7000	0.115775	0.115995	-0.022	0.241	0.8522	0.9839	no signal
OLMo-2	`keep8`@45000	0.117271	0.114046	$+0.322$	0.370	0.0856	0.2185	no signal
OLMo-2	`keep8`@121000	0.115442	0.113152	$+0.229$	0.446	0.3122	0.5620	no signal
Qwen3	`k8`, unhealed	0.107796	0.109185	-0.139	0.162	0.0964	0.2185	no signal
Qwen3	intact	0.461104	0.112212	$+34.889$	0.844	0.0001	0.0005	item-level signal
OLMo-2	intact	0.271858	0.112324	$+15.953$	0.761	0.0001	0.0005	item-level signal
OLMo-2	`keep10`@83500	0.112450	0.111508	$+0.094$	0.481	0.6966	0.8623	no signal
OLMo-2	`keep12`@124000	0.113115	0.112219	$+0.090$	0.471	0.7026	0.8623	no signal
OLMo-2	`keep14`@200000	0.119847	0.113766	$+0.608$	0.500	0.0172	0.0663	trace signal
OLMo-2	`shortgpt16`@200000	0.153341	0.112797	$+4.054$	0.575	0.0001	0.0005	item-level signal
Qwen3	`k10`, unhealed	0.118185	0.116155	$+0.203$	0.228	0.0804	0.2185	no signal
Qwen3	`k12`, unhealed	0.115027	0.114880	$+0.015$	0.379	0.9432	0.9968	no signal
Qwen3	`k14`, unhealed	0.118933	0.116263	$+0.267$	0.188	0.0066	0.0297	trace signal
Llama-2	intact	0.131981	0.112392	$+1.959$	0.525	0.0001	0.0005	anchor blocked
Llama-2	`k8`, unhealed	0.107463	0.111620	-0.416	0.495	0.1002	0.2185	no signal
Llama-2	`k10`, unhealed	0.113198	0.114575	-0.138	0.356	0.4510	0.7163	no signal
Llama-2	`k12`, unhealed	0.115858	0.116528	-0.067	0.081	0.1052	0.2185	no signal
Llama-2	`k14`, unhealed	0.116772	0.116408	$+0.036$	0.085	0.3950	0.6666	no signal
Llama-3	intact	0.329205	0.112014	$+21.719$	0.813	0.0001	0.0005	item-level signal
Llama-3	`k8`, unhealed	0.113531	0.115399	-0.187	0.351	0.2894	0.5581	no signal
Llama-3	`k10`, unhealed	0.111951	0.112269	-0.032	0.366	0.8746	0.9839	no signal
Llama-3	`k12`, unhealed	0.111868	0.111851	$+0.002$	0.514	0.9968	0.9968	no signal
Llama-3	`k14`, unhealed	0.114694	0.110945	$+0.375$	0.446	0.0970	0.2185	no signal

Table 7: Complete 27-cell read-out v2, with Benjamini–Hochberg q-values over exactly these 27 cells. Deltas and half-widths are percentage points. "Anchor blocked" denotes absolute item-level signal but no admissible relative-recovery claim. Numbers are emitted from the within-stratum permutation-null record computed on the per-item prediction files (evidence E-D; see the reproducibility statement for the artifact path). q_{BH} is the step-up value $\min_{j \geq i} mp_{(j)}/j$ with $m = 27$, enforced monotone; six cells have $q_{\text{BH}} \leq 0.05$ and five clear Bonferroni at $\alpha/27 = 0.001852$. Emitted by `gate_bh_27cell_consistency.py`, which reads this table as its only input.

A.10 REJECTED V2 ALTERNATIVES

Alternative	Cell	Null/statistic (pp)	Comparator (pp)	Measured reason for rejection
Own-modal null	Qwen3 `k8`	$+1.230$ vs always-E	v2 -0.139	Over-credits by 1.37 pp and flips the sign
Content interface	Qwen3 heal@7000	content -3.610	letter -0.022	More negative under the same permutation null
Content interface	Qwen3 `k8`	content -1.076	letter -0.139	More negative under the same permutation null
Content interface	OLMo-2 `keep8`@121000	content -0.086	letter $+0.229$	Does not rescue the damaged readout
Entropy/modal share	Llama-3 `k12`	$\Delta_{\text{perm}} = +0.002$	$p = 0.997$	Modal share 0.339 and entropy 0.614 still contain no signal

Table 8: Measured reasons for rejecting simpler v2 alternatives. Numbers are the same-cell comparisons recorded alongside the adopted permutation null (evidence E-D; see the reproducibility statement for the artifact path).

A.11 Integrity repairs

Defect	Consequence	Repair/hardening	Measured impact
Bootstrap zero atom counted in both tails	Illegal $p = 1.042$	Two-sided mid-p splits zero mass evenly; $p \leq 1$ structurally	0/24 verdicts changed; 0/30 crossed 0.05; non-p fields byte-identical
Tokenizer-specific left truncation	10/15 first-launch cells lost part of labelled option body	Raise cap to 2048; `n_trunc` becomes a hard per-shard assertion	0/14 verdicts changed; maximum 0.0083 pp; all eight argmax flips on affected items, 0 elsewhere
Llama-2 KV-cache OOM	5/8 shards failed; partial 3/8 result unavailable	Merge guard refused output; `use_cache=False` for teacher-forced pass	Peak memory 94 GiB to 41–50 GiB; intact cell recovered

Table 9: Integrity defects and repairs. Numbers are from the bootstrap mid-p repair record and the truncation/OOM re-launch audit of the first cross-family MMLU-Pro run (evidence E-E and E-I; see the reproducibility statement for the artifact paths).

A.12 Generated construct-null manifest

Table 10 restates the construct nulls of Table 1 as a *generated* artefact, and additionally retains the superseded legality-blind p column beside the corrected one. It is written by the manifest emitter directly from the balanced-null calibration record (evidence E-CAL), the single machine-readable source for these nine rows, and the emitter contains no numeric literal for any floor or chance line. The emitter refuses to write on any inconsistency: it checks that each stored percentage-point gap equals $100\,(\text{Floor} - \text{Chance})$, that each floor is a label count divided by n, that no floor falls below $1/k$, that each row's balanced-null verdict agrees both with its p-value and with the file's own partition of the constructs, that the corrected record agrees with the record it supersedes on every quantity the correction does not touch — n, k, floor, chance and gap — so that a "correction" cannot silently move a floor, and that the row identities agree with a second, independently recorded evidence file. It also re-parses every number in the hand-authored Table 1 and in its own output and re-verifies each against the source at the precision that table chose. A companion checker extends the same tolerance-based comparison to the prose. Both exit non-zero on any mismatch, so a typed number that drifts from the evidence becomes a build failure rather than a reading error. Both are named with their repository paths in Table 12.

Construct	n	k	Chance	Floor	Gap (pp)	p under null: blind	p under null: legal	Verdict
MMLU-Pro letter, naive	12032	10	0.100000	0.116606	+1.661	$< 10^{-5}$	0.083	inside estimator noise
MMLU-Pro letter, item-avg.	12032	10	0.110877	0.116606	+0.573	$< 10^{-5}$	0.083	inside estimator noise
MMLU letter	14042	4	0.250000	0.268908	+1.891	$< 10^{-5}$	$< 10^{-5}$	above balanced null
OpenBookQA letter	500	4	0.250000	0.276000	+2.600	0.383	0.384	inside estimator noise
ARC-Easy letter	2376	4	0.250161	0.266414	+1.625	0.140	0.136	inside estimator noise
ARC-Challenge letter	1172	4	0.250156	0.265358	+1.520	0.453	0.448	inside estimator noise
CommonsenseQA letter	1221	5	0.200000	0.208845	+0.884	0.853	0.853	inside estimator noise
PIQA letter	1838	2	0.500000	0.504897	+0.490	0.658	0.691	inside estimator noise
BoolQ	3270	2	0.500000	0.621713	+12.171	$< 10^{-5}$	$< 10^{-5}$	above balanced null

Table 10: **Construct-null manifest (generated, not transcribed), with the legality-blind null retained for contrast.** Every value is emitted programmatically from the legality-aware calibration record (evidence E-CAL) by the manifest emitter; no number in it is typed by hand. k is the nominal option count and Chance is the chance line used for that row, so the two MMLU-Pro rows differ only in whether chance is naive $1/10$ or the item average $\mathrm{mean}(1/\texttt{n\_opt})$. Gap is $100\,(\text{Floor} - \text{Chance})$, recomputed here rather than read. The two p columns are both $\Pr(\hat{f} \geq \text{Floor})$ under a balanced multinomial null, and differ only in whether that null is *realizable*. **blind** draws every gold letter uniformly over all k letters, which is the calibration this paper originally reported; for the 4 rows whose option count is not constant that assignment lies outside the support of any legal labelling, because an item with four options cannot have gold letter J. **legal** holds the observed `n_opt` histogram fixed and draws each item's gold letter uniformly among *its own* legal letters. Where `n_opt` is constant the two nulls are the same distribution and the columns agree by construction, which is itself a check on the emitter. Verdict follows the **legal** column. We keep the superseded column rather than deleting it because the incompatibility it illustrates is the paper's own thesis applied to the paper: a null must be compatible with the construct it calibrates, and the largest construct here is where we failed that test first. Provenance, from the same file: `schema_version` 1.0.0, `seed` 20260814, `n_draws` 1000000, `sha256 5ee8fa39d5fb`, superseding `sha256 275112623d05`. Method: `legality-aware balanced null: hold the observed n_opt histogram fixed and draw each item's gold letter uniformly among its own n_opt legal letters; f_const = max_L of the k sampled marginals; p = P(max_L >= observed floor).`

A.13 What we claim and what we retract

#	Claim previously considered	Current status	Manuscript-safe form
1	Letter behavior is a family-general step function or sharp phase transition.	Retracted	Report per-family descriptions only.
2	Damage turns the letter interface into a constant predictor.	Narrowed	Damage drives the letter score to or below its best-constant floor; modal share is separate.
3	Exact ties are the family-general causal mechanism.	Retracted	Precision changes ties and argmaxes but not accuracy; mechanisms are family-specific.
4	Letter MC is generally an unreliable instrument.	Retracted	Intact models can favor letter or content; calibrate each reported construct.
5	Accuracy-versus-normalized-accuracy length sensitivity is our finding.	Preempted	Attribute generic over-correction and tokenizer dependence to Oostermeijer; our result is floor inheritance.
6	Content is within ± 3 pp of letter on every damaged arm.	Re-scoped	False off MMLU; ARC-Easy reaches $+38.76$ pp.
7	Damaged letter scores are significantly below floor across small non-MMLU benchmarks.	Retracted	Only ARC-Easy `keep10` in OLMo-2; accompany other nulls with power.
8	`k14` is the last arm above floor in every family.	Retracted	Descriptive only for healed OLMo-2; other ladders are cliff-like and regime-confounded.
9	A pooled floor verdict can be read per benchmark.	Prohibited	Pool only per-item deviations and label the aggregate scope.
10	A content floor is a dataset property.	Narrowed	Token floors require dataset, convention, unit, and tokenizer; letter floors are dataset properties.
11	Larger BPE vocabularies create more option-length ties.	Retracted	Tokenizer dependence stands; the vocabulary-size mechanism is non-monotone.
12	Significant below-floor behavior is MMLU-specific.	Retracted and reinterpreted	MMLU-Pro has v1 below-floor cells, but v2 shows no negative item-level signal.
13	Damage universally drives letter to or below floor.	Narrowed	The honest MMLU-Pro count is 15/17; Qwen3 `k14` is a real but immaterial exception and `shortgpt16` clears its floor decisively.
14	First-launch cross-family MMLU-Pro numbers are usable.	Retracted	Use only the 2048-token corrected launch and canonical v2 evidence.
15	Clearing the floor certifies validity.	Retracted	Clearing is necessary, not sufficient; failure disqualifies, success does not certify.
16	Deprecated numeric claims remain usable.	Prohibited	Do not resurrect 4.8×, 0.2822, 58/91, MMLU 0.25, BoolQ 0.50, or 45.74 pp.

Table 11: Flat current-status ledger. The 16 rows and their status are the retraction/narrowing ledger maintained with the claim-to-evidence map, whose retracted-claim rows correspond one-to-one with the rows here (evidence E-K; see the reproducibility statement for the artifact path).

A.14 Reporting checklist

For each reported multiple-choice construct, we recommend the following minimal record:

1. Define the input-blind predictor family and report the best member on the evaluated item set.
2. Report the score, null value, paired effect, confidence interval, and test.
3. For content length nulls, print tie convention, length unit, and tokenizer. If option count varies, print the definition of chance.
4. Establish that the experiment could resolve the effect under discussion; otherwise describe the result as a non-observation.
5. Keep arm-independent interface validity separate from arm-conditional item-level information.
6. Treat a floor pass as necessary, not sufficient.

A.15 Evidence provenance

The construct floors, MMLU-Pro cell table, read-out v2 values, fp32 comparison, small-benchmark power analysis, integrity repairs, and claim ledger are transcribed from a single machine-extracted evidence record, whose lettered sections are the identifiers E-A through E-K used in the table captions. The character-versus-token table additionally uses that record's August 14 addendum, which records the raw-parquet reconstruction and 12/12 self-test. No healed Qwen3 step-121000 result is included.

Table 12 resolves every evidence identifier used in a caption to the file that carries it in the released artifact, together with the emitters and checkers named above. Captions cite the identifier rather than the path so that the claim remains legible without the artifact in hand; this table is the one place where the identifiers become literal paths, so a reader who does have the artifact can follow any caption to its source. Paths are relative to the repository's `paperC/` directory.

Identifier	Artifact path (relative to paperC/)
E-A	tcodex_out/EVIDENCE_PACK.md, Section A, re-derived from evidence/SECOND_MC_BENCHMARK_VERDICT.md
E-B	tcodex_out/EVIDENCE_PACK.md, Section B; per-cell records in evidence/mmlu_scale_power/ and evidence/POWER_WALL_VERDICT.md
E-D	evidence/heal_readout_v2_permutation_null.json (tcodex_out/EVIDENCE_PACK.md, Section D)
E-E	evidence/R7_BOOTSTRAP_P_FIX.md (tcodex_out/EVIDENCE_PACK.md, Section E)
E-F	evidence/second_mc_benchmark/ and evidence/SECOND_MC_BENCHMARK_VERDICT.md (tcodex_out/EVIDENCE_PACK.md, Section F)
E-H	evidence/construct_nulls_length_unit.json (tcodex_out/EVIDENCE_PACK.md, Section H)
E-I	evidence/mmlu_scale_power/trunc_fix_before_after.json and evidence/mmlu_scale_power/trunc_audit_1536.json, emitted by code/mmlu_pro_trunc_fix_compare.py and code/mmlu_pro_trunc_audit.py (tcodex_out/EVIDENCE_PACK.md, Section I)
E-K	tcodex_out/EVIDENCE_PACK.md, Section K; row-by-row map in evidence/claim_evidence_map.tsv
E-CAL	evidence/construct_nulls_legality_aware.json, which supersedes evidence/floor_winners_curse_calibration.json and names it, with its sha256, in its own supersedes field
Emitter	code/emit_tab_construct_nulls.py
Null recomputation	code/recompute_legality_aware_nulls.py
Prose checker	code/check_prose_vs_evidence.py

Table 12: Evidence identifiers used in the captions, resolved to artifact paths. The identifiers are stable; the paths are where those records live in the released repository.